Model Fit

Money Ballers

2025-11-22

```r
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed,
               knitr, jtools, tidyverse)

options(scipen = 999)

# Set theme for plots
theme_set(theme_minimal())
```

```r
# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"

# Reading data
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))
# dat <- read_parquet("player_df.parquet")
```

Table 1: Data Structure for fixed effects, except year as it is a random slope

Variable Type	**Variable**	Centering Strategy/ Form Factor	Justification
Time	Season Year	Min-Centering (+ Quadratic)	Establishes a meaningful starting intercept (2013 =0)
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation

Variable Type	**Variable**	Centering Strategy/ Form Factor	Justification
	Foreign Players %	Grand-Mean Centering	Moves baseline to “Average Club” (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect → Prime → Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps “Non-Citizen” as clear reference group. Separates individual status from club composition (`foreign_players`)
	Region of origin	Reference group, Europe	Control variable for where player is from
Player Performance	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player

Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```r
player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
    Region = fct_relevel(Region,  "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year),
    season_year_c_sq = season_year_c^2, # quadratic
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    age_c_sq = age_c^2,
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering
  group_by(season_year_c) |>
  mutate(uefa_coefficient = uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
         minutes_played = (minutes_played - mean(minutes_played, na.rm=TRUE)) /
           sd(minutes_played, na.rm = TRUE)) |>
  # position centering
  group_by(position) |>
  mutate(goals = goals - mean(goals, na.rm=TRUE),
         assists = assists - mean(assists, na.rm=TRUE)) |>
  ungroup()
```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

```r
# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()
```

Variable	Missing Cases
country_of_birth	108
country_of_citizenship	103
height_in_cm	36
date_of_birth	2
Age	2

```
clean_player_df <- player_df |> drop_na()
```

Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
  - Model 2.1: Random intercepts + fixed effect of centered season_year
  - Model 2.2: Adding quadratic terms of season year
  - Model 2.3 Adding random slope of season_year_c
  - Model 2.4: Adding quadratic term of season_year as a random slope
- Adding level fixed effects
  - Establish whether age quadratic term is needed
    * Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played}
    * Model 3.2: Adding the quadratic term for age
  - Model 4: Adding club covariates {UEFA coefficients + Foreign Players}
- Compare model fit
- Model diagnostic
- Choose final model

Model 1, intercepts only

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <-  lmer(
  usd_revenue_2024_log ~
    # random effects
      (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.19
BIC	20132.62
Pseudo-R^2 (fixed effects)	0.00
Pseudo-R^2 (total)	0.78

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.82	0.07	228.90	26.99	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.32
club_id	(Intercept)	0.28
Residual		0.72

- We can see that player_id accounts for about 75% of market value while club_id accounts for 3%.

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.03

Model 2.1: Adding fixed effect of year

```
Model_2.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_2.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.80
BIC	20087.09
Pseudo-R^2 (fixed effects)	0.00
Pseudo-R^2 (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.99	0.07	221.31	32.91	0.00
season_year_c	-0.03	0.00	-8.11	6524.03	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
      (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_2.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20046.98
BIC	20088.12
Pseudo-R^2 (fixed effects)	0.01
Pseudo-R^2 (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.92	0.07	214.88	36.55	0.00
season_year_c	0.02	0.01	1.58	5756.86	0.11
season_year_c_sq	-0.00	0.00	-4.45	5650.50	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```
anova(Model_2.1, Model_2.2, test = "LRT")
```

```
Data: clean_player_df
Models:
Model_2.1: usd_revenue_2024_log ~ season_year_c + (1 | player_id) + (1 | club_id)
Model_2.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + (1 | player_id) + (1 | c
          npar   AIC   BIC logLik -2*log(L) Chisq Df  Pr(>Chisq)
Model_2.1    5 20040 20074 -10015     20030
Model_2.2    6 20022 20063 -10005     20010 19.79  1 0.000008646 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 2.3, Random slope

- To estimate a random slope `(1 + year_centered | player_id)`, the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the "player slope" from the "residual error," causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

```
Model_2.3 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
      (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_2.3)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19986.28
BIC	20041.13
Pseudo-R^2 (fixed effects)	0.01
Pseudo-R^2 (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.90	0.09	182.85	26.54	0.00
season_year_c	0.02	0.01	1.55	116.87	0.12
season_year_c_sq	-0.00	0.00	-4.66	5632.32	0.00

p values calculated using Satterthwaite d.f.

Model 2.4, Random slope + quadratic term

- We have a convergence problem, it seems that adding the quadratic term makes the model unidentifiable. We move forward with Models 2.2 and 2.3.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.35
club_id	season_year_c	0.04
Residual		0.70

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.05

```r
Model_2.4 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
      (1 | player_id) + (season_year_c + season_year_c_sq | club_id),
  data = clean_player_df, REML = TRUE,    control = strict_control)

summ(Model_2.4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19961.36
BIC	20036.78
Pseudo-R^2 (fixed effects)	0.01
Pseudo-R^2 (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.90	0.10	163.08	22.83	0.00
season_year_c	0.02	0.03	0.91	19.23	0.37
season_year_c_sq	-0.00	0.00	-2.34	19.32	0.03

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.40
club_id	season_year_c	0.10
club_id	season_year_c_sq	0.01
Residual		0.70

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.74
club_id	21	0.07

Determine if the random slope on season year is needed

```r
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  # Formula: -2 * (LL_reduced - LL_full)
  lrt_stat <- -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  # This tells us how many params we added
  df_reduced <- attr(logLik(reduced_model), "df")
  df_full    <- attr(logLik(full_model), "df")
  df_diff    <- df_full - df_reduced
```

```r
  # 4. Calculate P-Value based on the difference
  # If we added a slope + covariance, df_diff should be 2.
  # The mixture is 0.5 * ChiSq(1) + 0.5 * ChiSq(2)

  if (df_diff == 2) {
    p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(1) and ChiSq(2)"

  } else if (df_diff == 1) {
    # If we added a parameter with NO covariance (e.g. diag structure)
    p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
             0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

    msg <- "Mixture of ChiSq(0) and ChiSq(1)"

  } else {
    # Fallback for non-standard comparisons
    p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
    msg <- paste0("Standard ChiSq(", df_diff, ")")
  }

  # Output
  cat("LRT Statistic:", round(lrt_stat, 4), "\n")
  cat("Parameter Diff:", df_diff, "\n")
  cat("Distribution:  ", msg, "\n")
  cat("P-Value:       ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.2, Model_2.3)
```

```
LRT Statistic: 64.7002
Parameter Diff: 2
Distribution:   Mixture of ChiSq(1) and ChiSq(2)
P-Value:        < 0.0001
```

- We use a helper function to print the p-value for the LRT difference, and find a p-value $<.0001$ and we reject the Null, (with $.5 \times\ 2(1) + .5 \times\ 2(2$ mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_3.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17497.18
BIC	17661.75
Pseudo-R^2 (fixed effects)	0.28
Pseudo-R^2 (total)	0.80

Model 3.2. Adding age quadratic term

```
Model_3.2 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.43	0.10	162.82	43.55	0.00
season_year_c	0.07	0.01	5.63	138.62	0.00
season_year_c_sq	-0.01	0.00	-9.05	5743.42	0.00
height_c	0.01	0.00	3.68	1677.18	0.00
positionDefender	-0.43	0.06	-7.19	1657.38	0.00
positionGoalkeeper	-1.24	0.10	-12.70	1674.56	0.00
positionMidfield	-0.22	0.06	-3.77	1685.59	0.00
citizen_of_club_countryYes	-0.40	0.04	-10.59	5702.02	0.00
RegionEast Asia & Pacific	-0.20	0.21	-0.93	1742.83	0.35
RegionLatin America & Caribbean	0.25	0.06	3.88	1803.67	0.00
RegionMiddle East & North Africa	0.10	0.25	0.39	1705.35	0.70
RegionNorth America	0.01	0.22	0.03	1787.91	0.98
RegionSub-Saharan Africa	0.02	0.10	0.21	1740.66	0.83
age_c	0.00	0.00	0.06	2366.73	0.95
goals	0.01	0.00	5.01	5556.92	0.00
assists	0.01	0.00	3.43	5095.87	0.00
red_cards	-0.01	0.03	-0.50	4713.31	0.62
yellow_cards	0.00	0.00	1.15	5428.13	0.25
minutes_played	0.52	0.02	29.58	5625.44	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.94
club_id	(Intercept)	0.37
club_id	season_year_c	0.03
Residual		0.62

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.63
club_id	21	0.10

```r
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_3.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14607.01
BIC	14778.43
Pseudo-R^2 (fixed effects)	0.39
Pseudo-R^2 (total)	0.89

Fixed Effects

	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.75	0.08	216.63	81.02	0.00
season_year_c	0.10	0.01	10.59	285.23	0.00
season_year_c_sq	-0.01	0.00	-14.17	5394.55	0.00
height_c	0.01	0.00	2.64	1799.23	0.01
positionDefender	-0.40	0.06	-6.94	1785.15	0.00
positionGoalkeeper	-1.06	0.09	-11.18	1797.22	0.00
positionMidfield	-0.19	0.06	-3.24	1801.62	0.00
citizen_of_club_countryYes	-0.19	0.03	-6.05	6841.18	0.00
RegionEast Asia & Pacific	-0.45	0.20	-2.19	1836.81	0.03
RegionLatin America & Caribbean	0.19	0.06	3.09	1953.26	0.00
RegionMiddle East & North Africa	-0.08	0.24	-0.32	1814.54	0.75
RegionNorth America	-0.05	0.22	-0.23	1858.52	0.81
RegionSub-Saharan Africa	0.07	0.10	0.70	1849.10	0.49
age_c	0.02	0.00	5.43	2304.05	0.00
age_c_sq	-0.02	0.00	-61.76	6187.82	0.00
goals	0.00	0.00	1.88	5191.96	0.06
assists	0.01	0.00	3.41	4899.62	0.00
red_cards	0.01	0.02	0.64	4687.80	0.52
yellow_cards	-0.00	0.00	-0.99	5112.22	0.32
minutes_played	0.36	0.01	26.55	5267.38	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.95
club_id	(Intercept)	0.23
club_id	season_year_c	0.02
Residual		0.47

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.77
club_id	21	0.05

- It seems that we need the quadratic term for age in the model.

```
anova(Model_3.1, Model_3.2)
```

```
Data: clean_player_df
Models:
Model_3.1: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c
Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c
          npar   AIC   BIC  logLik -2*log(L)  Chisq Df            Pr(>Chisq)
Model_3.1   24 17388 17552 -8669.9     17340
Model_3.2   25 14478 14650 -7214.1     14428 2911.8  1 < 0.00000000000000022

Model_3.1
Model_3.2 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 4: Adding club covariates

- The model with club covariates underperforms compared to Model 3.2.

```r
Model_4 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    uefa_coefficient + foreign_players +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14622.02
BIC	14807.16
Pseudo-R^2 (fixed effects)	0.39
Pseudo-R^2 (total)	0.89

```r
anova(Model_3.2, Model_4)
```

```
Data: clean_player_df
Models:
Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + ci
Model_4: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + citi
          npar   AIC   BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
Model_3.2   25 14478 14650 -7214.1     14428
Model_4     27 14479 14664 -7212.4     14425 3.3897  2     0.1836
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.76	0.08	214.00	76.65	0.00
season_year_c	0.10	0.01	10.00	274.96	0.00
season_year_c_sq	-0.01	0.00	-13.37	5374.35	0.00
height_c	0.01	0.00	2.65	1796.69	0.01
positionDefender	-0.40	0.06	-6.95	1782.41	0.00
positionGoalkeeper	-1.06	0.09	-11.18	1794.71	0.00
positionMidfield	-0.19	0.06	-3.24	1798.93	0.00
citizen_of_club_countryYes	-0.19	0.03	-5.95	6851.81	0.00
RegionEast Asia & Pacific	-0.45	0.20	-2.19	1834.38	0.03
RegionLatin America & Caribbean	0.19	0.06	3.07	1951.10	0.00
RegionMiddle East & North Africa	-0.08	0.24	-0.33	1811.89	0.74
RegionNorth America	-0.05	0.22	-0.24	1856.11	0.81
RegionSub-Saharan Africa	0.07	0.10	0.68	1846.84	0.50
age_c	0.02	0.00	5.41	2303.32	0.00
age_c_sq	-0.02	0.00	-61.75	6198.23	0.00
goals	0.00	0.00	1.90	5187.13	0.06
assists	0.01	0.00	3.42	4895.10	0.00
red_cards	0.01	0.02	0.61	4682.99	0.54
yellow_cards	-0.00	0.00	-0.99	5107.27	0.32
minutes_played	0.36	0.01	26.56	5262.54	0.00
uefa_coefficient	-0.00	0.00	-1.16	4432.07	0.24
foreign_players	0.14	0.10	1.46	1851.74	0.14

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.95
club_id	(Intercept)	0.24
club_id	season_year_c	0.02
Residual		0.47

Bonus round, Model 5 testing 2 levels only for player data.

- In a two level model dropping the random intercept on club, which ultimetly drops the random slope, is a worst fit than model 3.2. It appears we need the random slope of year, but not necessirily the the club covariates.

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.77
club_id	21	0.05

```
Model_5 <-  lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id),
  data = clean_player_df, REML = TRUE,   control = strict_control)

summ(Model_5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14928.80
BIC	15079.65
Pseudo-R^2 (fixed effects)	0.38
Pseudo-R^2 (total)	0.89

```
anova(Model_3.2, Model_5)
```

```
Data: clean_player_df
Models:
Model_5: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + citi
Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + ci
          npar   AIC   BIC  logLik -2*log(L)  Chisq Df            Pr(>Chisq)
Model_5     22 14798 14949 -7377.1     14754
Model_3.2   25 14478 14650 -7214.1     14428 326.17  3 < 0.00000000000000022
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.78	0.06	275.62	2451.14	0.00
season_year_c	0.10	0.01	11.22	6862.95	0.00
season_year_c_sq	-0.01	0.00	-13.77	5394.07	0.00
height_c	0.01	0.00	2.29	1899.04	0.02
positionDefender	-0.37	0.06	-5.98	1902.41	0.00
positionGoalkeeper	-1.03	0.10	-10.25	1911.41	0.00
positionMidfield	-0.15	0.06	-2.42	1917.48	0.02
citizen_of_club_countryYes	-0.27	0.03	-8.59	7002.32	0.00
RegionEast Asia & Pacific	-0.48	0.22	-2.23	1942.91	0.03
RegionLatin America & Caribbean	0.09	0.06	1.41	2016.74	0.16
RegionMiddle East & North Africa	-0.16	0.25	-0.65	1929.81	0.51
RegionNorth America	-0.10	0.23	-0.42	1970.98	0.68
RegionSub-Saharan Africa	0.03	0.11	0.26	1949.84	0.79
age_c	0.02	0.00	5.04	2366.63	0.00
age_c_sq	-0.02	0.00	-63.69	6042.27	0.00
goals	0.00	0.00	1.59	5197.50	0.11
assists	0.01	0.00	3.47	4983.99	0.00
red_cards	-0.00	0.02	-0.04	4796.43	0.97
yellow_cards	-0.01	0.00	-2.91	5182.20	0.00
minutes_played	0.36	0.01	26.02	5278.50	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.02
Residual		0.47

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.82

```
Model_5
Model_3.2 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
export_summs(Model_2.3, Model_3.2, Model_4,
              digits=3, r.squared=FALSE,
             model.names = c("Model_2.3", "Model_3.2", "Model_4"))
```

```
final_model <- Model_3.2
```

Model Diagnostics

Linearity

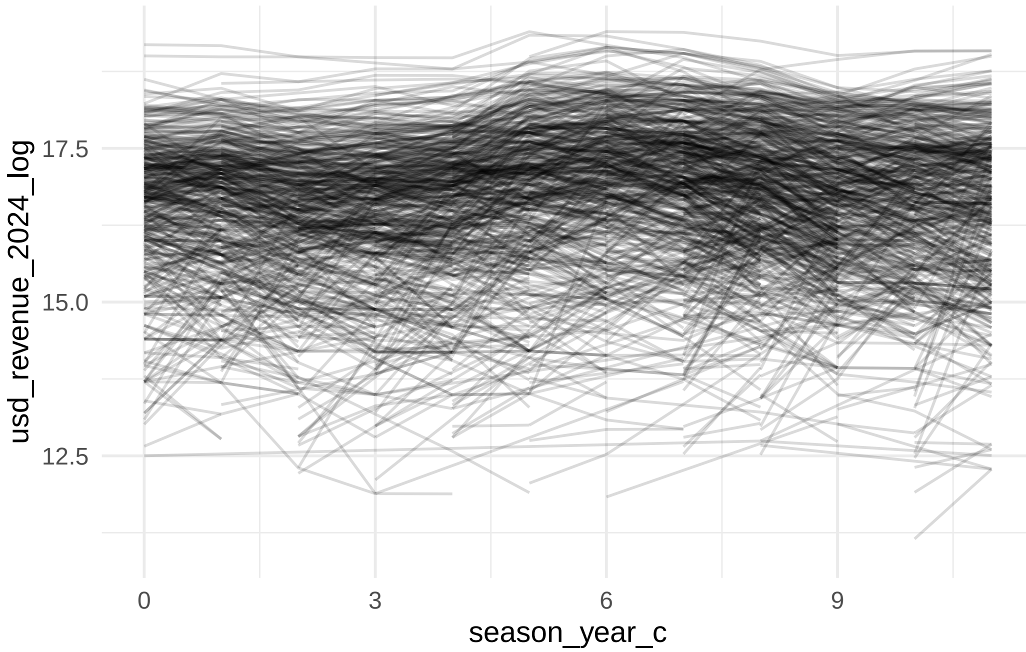
In the overall spaghetti plot, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (usd_revenue_2024_log) over centered season year (season_year_c). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern.

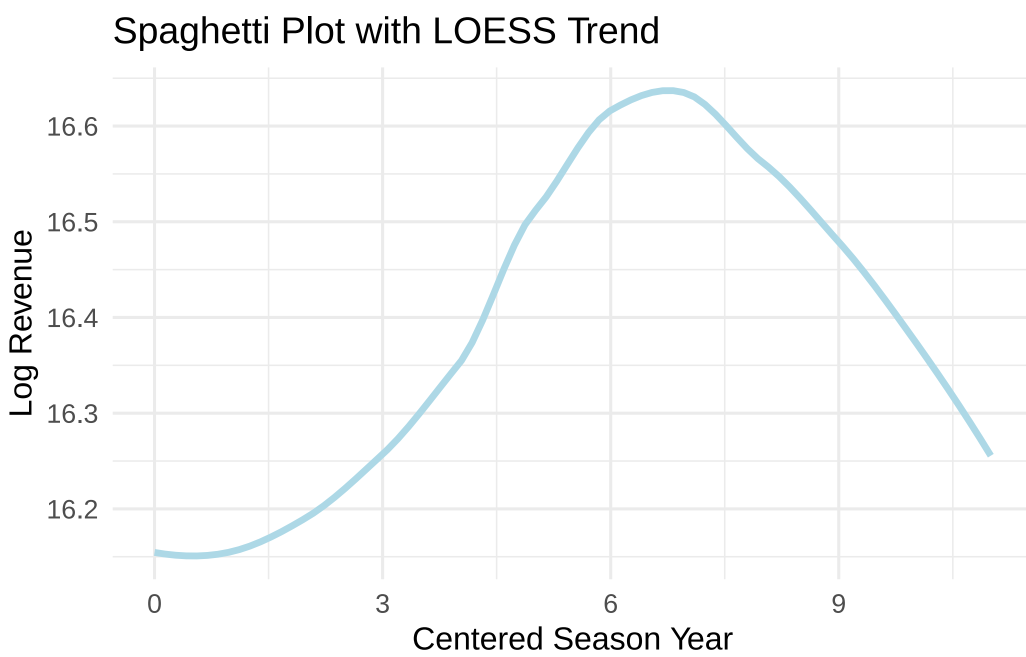
By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers.Thus, we cannot say the normal random effects assumption in the multilevel model are met.

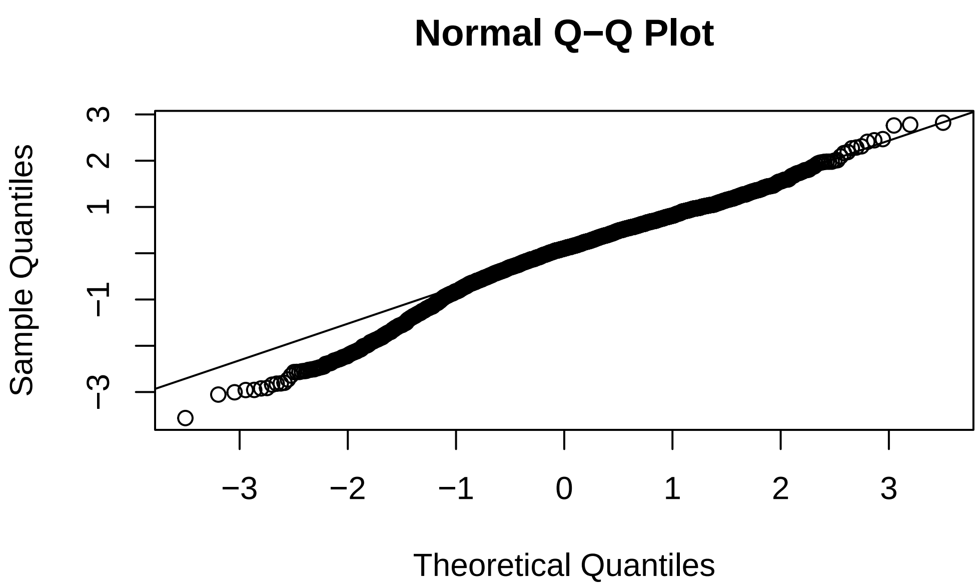
```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
              group = player_id),color = "black", alpha = 0.15)
```



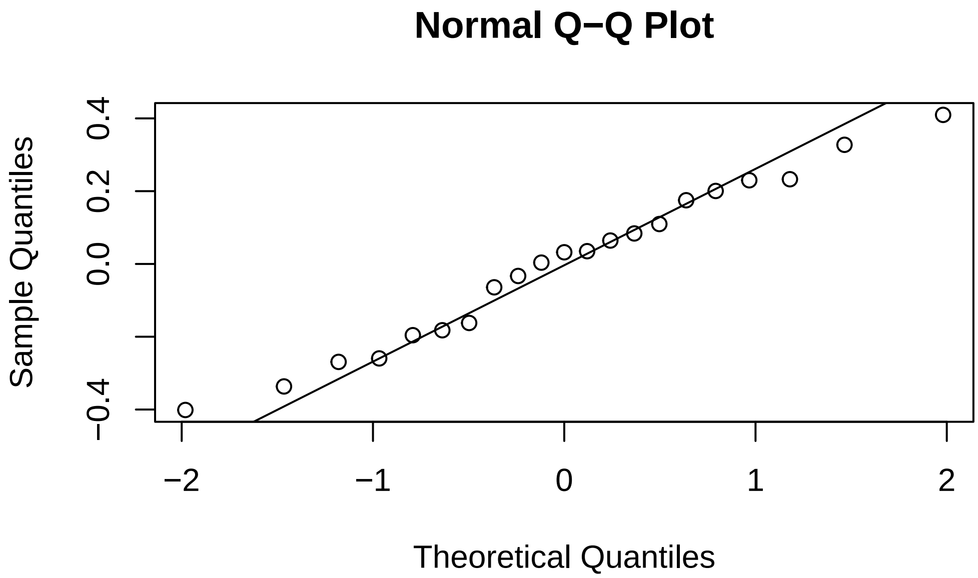

```
ggplot(clean_player_df, aes(x = season_year_c,
                            y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
              color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



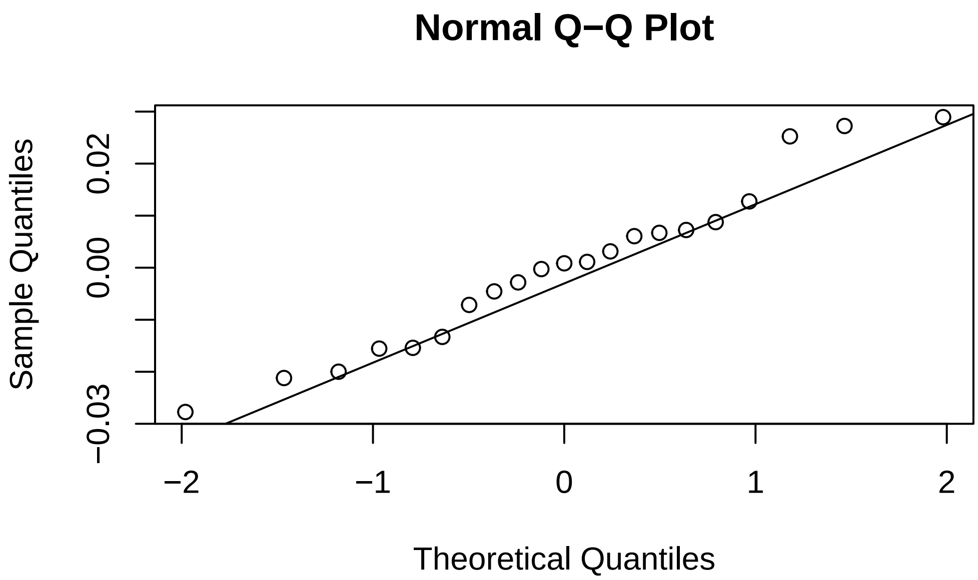

```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```



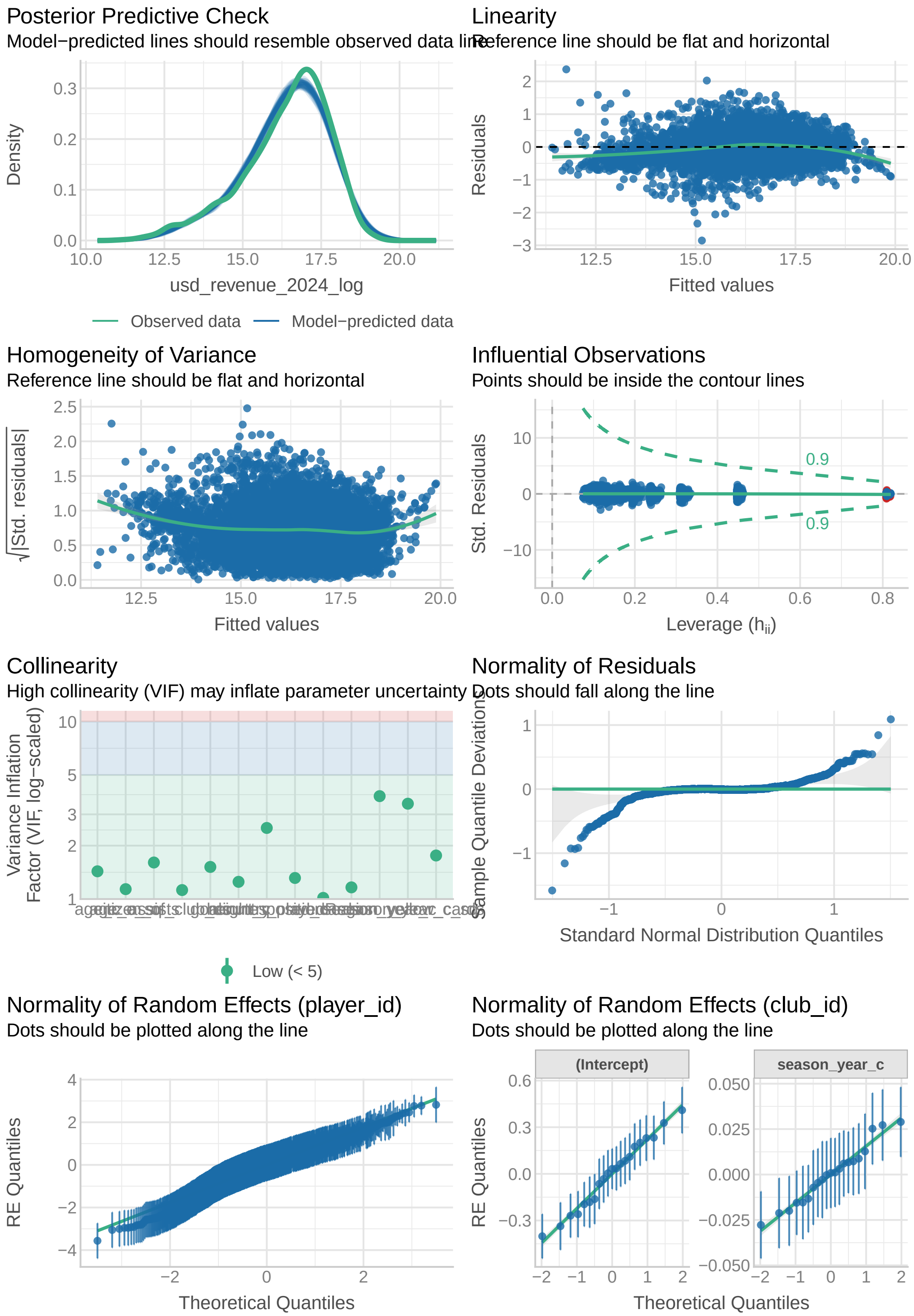
```r
qqnorm(ranef(final_model)$club_id[,1]) # group factor
qqline(ranef(final_model)$club_id[,1])
```




```r
qqnorm(ranef(final_model)$club_id[,2]) # group factor
qqline(ranef(final_model)$club_id[,2])
```




```
check_model(final_model)
```


Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
usd_revenue_2024_log
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (hii)
0.9
0.9
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
Standard Normal Distribution Quantiles
Normality of Random Effects (player_id)
Dots should be plotted along the line
RE Quantiles
Theoretical Quantiles
Normality of Random Effects (club_id)
Dots should be plotted along the line
(Intercept)
season_year_c
RE Quantiles
Theoretical Quantiles

Final Model Specification

To test our hypotheses regarding the determinants of player market value, we fitted a **cross-classified multilevel model** (Model 3.2). This structure accounts for the non-hierarchical nature of the data, where players (i) move between different clubs (j) over time (t).

The final model specification for the log-transformed market value, Y_{tij}, is formally expressed as:

$$
\begin{aligned}
\log(\text{Revenue})_{tij} = \beta_0 + & \underbrace{\beta_1(\text{Year}_t) + \beta_2(\text{Year}_t^2)}_{\text{Time Trends}} \\
& + \underbrace{\beta_3(\text{Height}_i - \bar{\text{Height}}) + \beta_4(\text{Citizen}_{ij}) + \beta_5(\text{Age}_{it} - \bar{\text{Age}}) + \beta_6(\text{Age}_{it}^2)}_{\text{Player Fixed Effects (Traits \& Demographics)}} \\
& + \underbrace{\sum_{p=1}^{P-1} \beta_{Pos,p}(\mathbb{I}_{\text{Position}_i = p}) + \sum_{r=1}^{R-1} \beta_{Reg,r}(\mathbb{I}_{\text{Region}_i = r})}_{\text{Categorical Controls}} \\
& + \underbrace{\beta_7\text{Goals}_{it} + \beta_8\text{Assists}_{it} + \beta_9\text{Red}_{it} + \beta_{10}\text{Yellow}_{it} + \beta_{11}(\text{Mins}_{it} - \bar{\text{Mins}}_{tj})}_{\text{Performance \& Status (Time-Varying)}} \\
& + \underbrace{u_{0i}}_{\text{Player RE}} + \underbrace{v_{0j} + v_{1j}(\text{Year}_t)}_{\text{Club RE}} + \epsilon_{tij}
\end{aligned}
$$

Where the random effects are distributed as:

$$
\begin{aligned}
u_{0i} &\sim N(0, \sigma^2_{\text{player}}) \\
\begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} &\sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2_{\text{club\_int}} & \sigma_{\text{cov}} \\ \sigma_{\text{cov}} & \sigma^2_{\text{club\_slope}} \end{pmatrix} \right) \\
\epsilon_{tij} &\sim N(0, \sigma^2_\epsilon)
\end{aligned}
$$

Interpretation of Results

Fixed Effects (Average Predictors)

- **Intercept ($\beta_0 = 16.75$):** The estimated log market value for a baseline player (Attacker from Europe/Central Asia) with average physical traits and career status at the start of the study period (2013).
- **Market Trajectory ($\beta_1 = 0.10, \beta_2 = -0.01$):** The significant positive linear term indicates the transfer market grew substantially over the decade. However, the significant negative quadratic term (β_2) reveals a **market deceleration**, where the rapid inflation of the early 2010s has slowed or plateaued in recent years.

- **Player Status ($\beta_{11} = 0.36$):** We centered *Minutes Played* relative to the specific club and year. The positive coefficient indicates that being a **"Starter"** (playing more minutes than the average teammate) is associated with a significant premium in market value, independent of the club's prestige.
- **Biological Age ($\beta_5 = 0.02, \beta_6 = -0.02$):** Using grand-mean centering, we observe a classic parabolic career arc. Value increases as a player moves from "Prospect" to "Prime," but the negative quadratic term confirms that value declines significantly as players enter their "Veteran" years.
- **Homegrown Effect ($\beta_4 = -0.19$):** The variable `citizen_of_club_country` serves as a proxy for domestic status. The negative coefficient implies a **"Foreign Premium"** (or domestic discount), where non-citizen players are valued approximately 19% higher than domestic players, likely reflecting the recruitment costs associated with importing talent.

Random Effects (Variance Components)

- **Player Heterogeneity ($\sigma_{player} = 0.95$):** The substantial random intercept for players (ICC $\approx$ 0.77) suggests that **reputation and brand** ("The Superstar Effect") drive the majority of market value variance, rather than just annual performance statistics.
- **Club Trajectories ($\sigma_{slope} = 0.02$):** The significant random slope for `season_year` confirms that clubs do not share a parallel growth path. Some clubs act as "Incubators" (accelerating player value faster than the market), while others remain stagnant.
- **The "Catch-Up" Effect ($r = -0.35$):** The negative correlation between the Club Intercept and Club Slope suggests that clubs starting with lower baseline values (smaller clubs) often see steeper relative growth rates in their assets compared to established super-clubs, likely due to their role in developing talent for resale.

	Model_2.3	Model_3.2	Model_4
(Intercept)	15.903 ***	16.752 ***	16.757 ***
	(0.087)	(0.077)	(0.078)
season_year_c	0.022	0.099 ***	0.096 ***
	(0.015)	(0.009)	(0.010)
season_year_c_sq	-0.005 ***	-0.010 ***	-0.009 ***
	(0.001)	(0.001)	(0.001)
height_c		0.010 **	0.010 **
		(0.004)	(0.004)
positionDefender		-0.400 ***	-0.400 ***
		(0.058)	(0.058)
positionGoalkeeper		-1.056 ***	-1.055 ***
		(0.094)	(0.094)
positionMidfield		-0.186 **	-0.186 **
		(0.058)	(0.057)
citizen_of_club_countryYes		-0.192 ***	-0.189 ***
		(0.032)	(0.032)
RegionEast Asia & Pacific		-0.446 *	-0.446 *
		(0.203)	(0.203)
RegionLatin America & Caribbean		0.188 **	0.186 **
		(0.061)	(0.061)
RegionMiddle East & North Africa		-0.076	-0.077
		(0.237)	(0.237)
RegionNorth America		-0.051	-0.052
		(0.216)	(0.216)
RegionSub-Saharan Africa		0.069	0.068
		(0.100)	(0.100)
age_c		0.020 ***	0.020 ***
		(0.004)	(0.004)
age_c_sq		-0.024 ***	-0.024 ***
		(0.000)	(0.000)
goals		0.004	0.004